Machine-Learning-Based Multidimensional Big Data Analytics over Clouds via Multi-Columnar Big OLAP Data Cube Compression

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***Abstract*—This paper proposes a new theory on combining innovative *Multidimensional Big Data Analytics* with well-known *Machine Learning* (ML) in order to magnify the expressive power and the accuracy of knowledge insights discovery from massive big datasets. At the level of *enabling technology*, with the goal of fully supporting this novel paradigm, the issue of *managing and mining big OLAP data cubes over Clouds* arises. Due to computational complexity requirements, the latter challenge is addressed by proposing an innovative solution for (1) representing big OLAP data cubes over Clouds via a *multi-column-based representation*, and (2) *compressing* the deriving multi-column representations for achieving the desired effectiveness and efficiency. This paper introduces the fundamental model of *Machine-Learning-Based Multidimensional Big Data Analytics*, along with a reference architecture implementing it.**

Keywords—Big Data, Big Data Analytics, Multidimensional Big Data Analytics, Machine Learning, Machine-Learning-Based Multidimensional Big Data Analytics, Big OLAP Cubes, Multi-Columnar Representation of Big OLAP Cubes, Multi-Columnar Big OLAP Cube Compression

I. Introduction

Nowadays, *OnLine Analytical Processing* (OLAP) and *Business Intelligence* (BI) have conquered the scene of decision-making systems research for decades (e.g., [1,2]), thus defining a wide collection of models, techniques, and algorithms whose main goal is that of effectively and efficiently representing, querying, and mining massive OLAP data cubes. With the advent of the modern Big Data era (e.g., [3-5]), Big OLAP Data Cubes arise (e.g., [6-8]), thus posing novel challenges to be faced-off by the research community. Big OLAP Data Cubes are mainly processed over Clouds due to tight high-performance computing requirements, thus opening the door to a large number of big data processing platforms from both the academic and industrial worlds.

Coupled with the problem of big OLAP data cubes, the new research line falling under the term "*Multidimensional Big Data Analytics*" appeared (e.g., [9-12]), with an important impact on the research community. Basically, this novel paradigm pursues the goal of engrafting multidimensional analysis into the whole big data analytics process, as to magnify the expressive power and the retrieved accuracy in the (final) decision-making process.

Indeed, *big data repositories* (e.g., [13-15]) contain data of *great value*. But, *how to extract these interesting patterns from such large-scale repositories?* The latter is a fundamental research question that is still open. Recent approaches that appeared in literature mostly follow the idea of applying classical *Data Mining* (DM) and *Machine Learning* (ML) algorithms to these repositories (e.g., [16-18]), by devising rather-innovative optimizations aimed at coupling with specific characteristics of big data, including *volume*, *velocity*, *variety,* and *veracity* [19].

Actually, even with the modern connection with emerging paradigms like Explainable Artificial Intelligence, which well-couples with ML (e.g., [20]), one of the most relevant limitations of the popular DM and ML approaches to big data analytics is still represented by the accuracy and expressive power of the process itself. This is just due to the intrinsic nature of big data repositories, which expose the typical 3V characteristics.

Despite the classical approach, thanks to the emerging of Cloud and high-performance technologies, nowadays a new paradigm is possible. This paradigm falls under the name of advanced ML structures over big data repositories, and its main goal is that of supporting meaningful Multidimensional Big Data Analytics. These "structures" are rather programs than tasks so that they incorporate ML procedures within high-level (program) controls whose main goal is that of magnifying the expressive power of the whole big data analytics process implemented as a collection of singleton big data analytics tasks. In turn, each task is implemented in terms of a proper advanced ML structure.

While some approaches are starting to follow this novel direction (e.g., [21-25]), the main idea is completely innovative, and it also has a direct connection with the research area predicating to support big data analytics over commodity hardware (e.g., [26]). Basically, the advanced ML programs run on top of massive big OLAP data cubes to extract actionable knowledge from them by also realizing multiple-step iterations. It should be noted that this model is completely alternative to the classical DM model, where the main discovery procedure is single-step.

The proposed model also needs to address another relevant problem. Indeed, today's enterprise analytics workloads have become computationally-intensive, with also complex sharing among scientific applications and computing applications,

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which further makes worse the target scenario. As a consequence, the overall amount of data and processing accessed and processed during the execution of these workloads is prohibitive, and, as a consequence, parallel executions are necessary in order to reduce the whole computational cost. Therefore, higher-level data-center architectures and scalable ML programs, with also parallel cores, are the most appropriate solutions for the setting we are investigating.

The so-depicted scenario opens the door to a plethora of relevant research challenges. First, the granularity of the data processing flow. When data are partitioned in chunks and parallel pieces of code are running, supported big data analytics can execute with a time granularity much higher than the time granularity of parallel cores. This temporal difference determines relevant practical drawbacks. Second, failures. Data/computational failures are always possible in such complex systems. Therefore, preventing failure events in big data systems is a very annoying problem to deal with. Third, openness and standardization. Big data systems for supporting ML structures are powerful systems. As a consequence, they are usually organized into sub-systems. When sub-systems are advocated, then hardware and software heterogeneities arise, and the openness and standardization of models, data, systems, and tools become a first-class requirement to deal with.

In this so-delineated context, the main research challenges again consist of the following issues: (*i*) effectively and efficiently representing big OLAP data cubes over Cloud architectures; (*ii*) effectively and efficiently querying Big OLAP data cubes over Cloud architectures; (*iii*) effectively and efficiently mining big OLAP data cubes over Cloud architectures using advanced ML structures. Following this main research challenge, recently a plethora of research proposals have appeared in literature. This initiative founds its roots in decades of research in database representation and query optimization (e.g., [27]). In this area, many approaches have been proposed across years. In this paper, we especially focus on data compression paradigm (e.g., [28]), which has animated and is continuing to do so the database research community first and, again, the big data research community at now.

Following these considerations, in this paper, we propose a novel approach of combining multidimensional big data analytics with machine learning. Our aim is to not only mitigate the challenges associated with accuracy but also enhance the overall expressive power of the big data analytics process. Specifically, we will delve into the intricacies of advanced ML methodologies by presenting how they can be performed over big data. As well as presenting an innovative approach for managing and representing big OLAP data cubes within Cloud-based environments via multi-column data cube compression.

Through this paper, we seek to present a forward-looking perspective on the convergence of multidimensional big data analytics and ML. Our proposed approach is designed to be both versatile and future-proof, accommodating emerging trends and technologies in the rapidly evolving landscape of data science.

The remaining part of this paper is organized as follows. Section II illustrates the definitions and examples of applying advanced ML techniques and algorithms to the analysis of complex, multidimensional big data. In Section III, we introduce and describe the anatomy and functionalities of the reference architecture for supporting ML structures over big data repositories. After that, in Section IV, we present the complementary architecture that deals with the issue of managing and mining big OLAP data cubes in Cloud environments. Finally, Section V provides conclusions of this research as well as sketching possible ways for future work. This work extends significantly two previous experiences [56,57], where we initially proposed the embryonic ideas of our research.

II. Advanced Machine Learning Structures over Big Data Repositories: Definitions and Examples

In this Section, we provide an in-depth exploration of *advanced ML techniques and algorithms* applied to the analysis of complex, multidimensional datasets stored within big data repositories. It aims to elucidate these sophisticated techniques by offering clear definitions and demonstrating their practical usage, benefits, and limitations within this investigated context.

Advanced ML structures over big data repositories refer to the application of sophisticated ML algorithms over multidimensional big data in order to extract useful knowledge and valuable insights. Indeed, these advanced structures are playing a pivotal role in harnessing the potential of massive data repositories; they are being applied across various tasks ranging from *image recognition* and *recommendation systems* to *anomaly detection*, *risk assessment*, and *predictive analytics*.

Nevertheless, before applying these algorithms, the ML structure defines the domain on which the ML models are built. Particularly, by specifying (*i*) the data source view, which defines the subset of data that will be used for training and testing; (*ii*) the number and type of columns, which describe the data contained by the data source; (*iii*) partitioning the data into training and testing sets, which allow to specify that a certain percentage of data is held out for testing and the rest is used for training. This particularity admits that it is no longer necessary to separate the raw data in advance. In addition, after defining the domain, the next step consists of choosing the right ML algorithms and training the corresponding model over the training set, as well as evaluating its performance on the testing set.

As an important advanced ML structure that can be used in this context, *Neural Networks* (NN) are inspired by the structure and functioning of biological neural networks. They are capable of automatically extracting relevant features from data and modeling complex, non-linear relationships, making them particularly well-suited for complex problems. Additionally, neural networks are versatile and can be used for various tasks; as a consequence, they play a significant role in the analysis of multidimensional big data repositories by offering several advantages, such as:

- *Complex Data Structures*: big data repositories often contain diverse and complex data types, such as images, text, and time-series data. Neural networks can process and analyze such data, making them a versatile choice for various data types;
- *Features Extraction*: they can automatically extract relevant features from vast datasets, reducing the need for manual feature engineering. This is especially beneficial when dealing with unstructured or semi-structured data;
- *Pattern Recognition*: they are suitable for recognizing intricate patterns within large datasets. This ability is particularly valuable in big data analytics, where

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traditional statistical methods may struggle to uncover hidden insights;

Predictive Modeling: neural networks excel at predictive modeling. They can make accurate predictions and classifications, which is crucial in tasks like demand forecasting, fraud detection, and recommendation systems within large datasets – big data repositories often contain diverse and complex data types, such as images, text, and time-series data. Neural networks can process and analyze such data, making them a versatile choice for various data types;

- *Anomaly Detection*: neural networks can help to identify anomalies or outliers in large datasets, which is essential for various tasks (e.g., fraud detection, network security, quality control, etc.);
- *Real-Time Processing*: in some applications, such as real-time monitoring or instant decision-making, neural networks can process data quickly and efficiently. This is crucial when dealing with data streams generated in real-time.

Despite these advantages, neural networks still have some challenges when coupled with big data repositories, such as *computational complexity*, as they can be computationally expensive to be trained, especially over very large datasets. As a solution to address these drawbacks, *TensorFlow* has gained a significant focus from researchers in the active literature due to the fact that it supports both large-scale training and inference. *TensorFlow* efficiently uses hundreds of powerful (*GPU-enabled*) servers for fast training and runs models for inference on various platforms, ranging from large distributed clusters, down to running locally on mobile devices [29].

Furthermore, *Logistic Regression* is another popular ML algorithm that can be employed in this situation. It represents a statistical model and a fundamental algorithm in ML used for binary classification tasks. Unlike linear regression, which predicts continuous values, logistic regression predicts the probability of an input belonging to a specific class. Logistic regression is particularly suitable for big data repositories because it can handle large amounts of data and complex relationships between variables. It is also relatively efficient to train and evaluate models on big multidimensional data. Logistic regression is utilized in several applications within the context of multidimensional big data (e.g., risk assessment, marketing, medical diagnosis, and so forth).

Moreover, the usage of logistic regression over big data repositories can lead to various benefits, like (*i*) *scalability and efficiency* for their ability to effectively and efficiently handle very large and complicated datasets; (*ii*) *interpretability*, logistic regression models are relatively easy to interpret and understand. However, although these benefits, there are still several challenges and issues, such as:

- *Features Extraction*: it is important to carefully select and engineer the features that are used to train a logistic regression model. This can be a challenging task, especially when dealing with big multidimensional data;
- *Hyperparameter Tuning*: these models have several hyperparameters that can be tuned to improve their performance. This can be a time-consuming and computationally expensive process;
- *Overfitting*: logistic regression models can overfit the training data, which means that they may not perform well on new data. It is important to use cross-validation and other techniques to avoid this issue.

However, logistic regression is a powerful and simple yet effective algorithm for extracting insights from multidimensional big data repositories and can be used to solve a wide range of problems within this context.

In summary, one significant novelty of this approach consists of coupling *advanced ML structures with multidimensional modeling*, which falls under the recent innovation of *multidimensional big data analytics proposals* (e.g., [12]). This novel idea predicates the marriage between expressive multidimensional models (e.g., [30,31]) and powerful ML algorithms, with the final goal of enhancing the expressive power and accuracy of the whole knowledge discovery process over very-large big data repositories. As it can be noticed by the contributions of this Section, decades of ML research represent the natural humus for the novel research area we investigate in this paper.

III. A Reference Architecture for Supporting Machine Learning Structures over Big Data Repositories: Anatomy and Main Functionalities

In this Section, we introduce the reference architecture for supporting ML structures over big data repositories, where we provide an overview of how ML structures and models can be employed on top of traditional big data processes and big data analytics tasks in order to extract knowledge from different types of huge and heterogeneous data repositories. The use, management, and combination of such types of structures allow for providing analytical insights and a better understanding of these large amounts of data and, therefore, improve the decision-making process. Fig. 1 depicts in detail our proposed architecture.

As shown in Fig. 1, our proposed architecture encompasses several functional components that work collaboratively to enable advanced ML structures over big data repositories, which we describe as follows:

1) Big Data Processing Layer: it is the first phase of the entire process of big data analytics over big data repositories using ML structures, and it is based on two main steps (i.e., *Data Preparation* and *Data Pre-Processing*) where:

a) The first stage of developing a ML model is data preparation, which selects the best data samples for modeling after obtaining an overview of the data. The main tasks of this step aim to (*i*) extract the dataset from historical databases and other big data sources, analyze its structure, and select relevant data; (*ii*) guarantee information extraction efficiency and analyze data natures or characteristics, such as non-Gaussianity, linear/nonlinear relationships among various attributes, time-series correlations, etc.; (*iii*) select samples and attributes [32]. It is dependent upon the type of model we wish to create and its main task.

Following that, data pre-processing must be done to improve data quality, and some appropriate data transformations may be required to increase efficiency in data modeling. This includes (*i*) carefully checking the data and eliminating inconsistencies; (*ii*) removing outliers and gross errors from the modeling dataset, as they will otherwise significantly deteriorate the performance of the ML model [33]; and (*iii*) there may be some missing data in the prepared dataset, this must be addressed by missing value estimation, *Bayesian inference*, etc. [34,35]. Since various models or learning tasks may have requirements on different scales for distinct data properties, data preparation needs to be carefully considered. Various techniques for data transformation and scaling can

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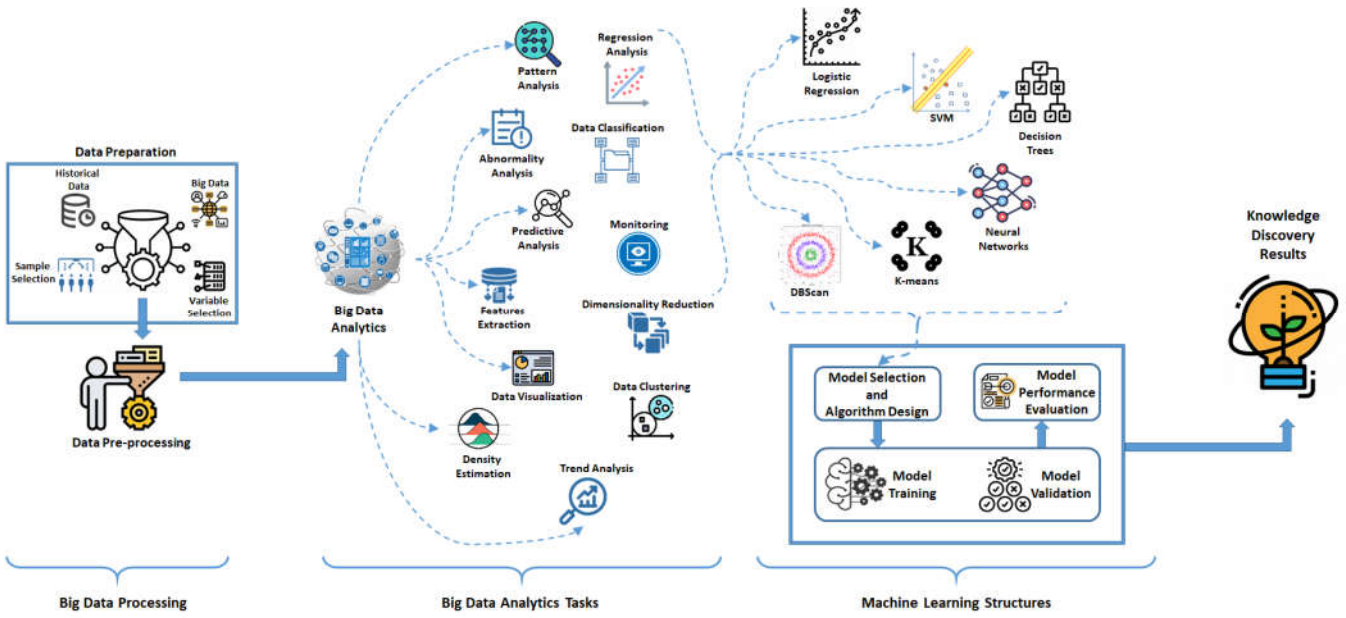

Fig. 1. Reference Architecture for Supporting ML Structures over Big Data Repositories

improve the performance of the data model in this scenario.

2) Big Data Analytics Layer: in this phase, several big data analytical tasks are carried out, such as data clustering, dimensionality reduction, data visualization, trend analysis, monitoring and fault diagnosis, fault classification, prediction, and so on. Some of these examples are described as follows:

a) Appropriate statistics can be created for online monitoring of various circumstances in order to monitor and diagnose faults. For instance, multivariate statistical monitoring techniques are typically used to create monitoring statistics. It is possible to distinguish between good and poor situations by setting control limits for these statistics. A fault warning indicating abnormal activity is triggered if the statistical values surpass the applicable control limits. Various techniques for fault diagnosis can be used to identify the fault's main causes and produce assessment reports, which will allow for later optimizations or enhancements. Problems with data and fault categorization might arise from a variety of anomalous occurrences or faults that happen but cannot be diagnosed [36]. ML offered several helpful techniques for fault classification modeling, which are described in Section II.

b) Clustering analysis may be performed in both offline and online phases to investigate data patterns in a dataset (e.g., *how many clusters exist within the entire dataset*?). Several clusters can be identified during the offline phase. Depending on what particular big data analytics can be performed for each cluster, the results of clustering can be used to analyze the pattern or mode of the dataset. As a result, several improvements or decisions can be made. Patterns in new data samples may be found and identified in the online stage. As a result, the corresponding data model may be used to extract information and analyze new data, significantly increasing the effectiveness of online data analytics.

c) Due to the characteristics of big data and its high dimensionality, and as data attributes have a significant correlation with one another, which makes data visualization difficult, carrying out analytics on the original multidimensional data is challenging. For handling such massive volumes of data in an efficient manner, dimensionality reduction is therefore required. By doing so, the computing complexity of the subsequent data analytics methods is reduced, and important information may be extracted.

d) Prediction of key performance indicators is essential since, unlike normal attributes, they are exceedingly difficult to measure in real-time online, which causes major delays. Nonetheless, these critical performance metrics may be estimated online through the use of predictive modeling approaches. Predictive data models may be developed and used for online prediction of new data samples, yielding real-time predictions, by identifying the regression correlations between the regular data attributes and those key performance indicators. These regression-based predictive big data analytics models may be used in a variety of scenarios, including estimating quality, energy consumption, assessing environmental performance, calculating economic performance, and more.

3) Machine Learning Structures Layer: in this layer, ML models are selected, trained, validated, and tested to be used in different analytical tasks using different ML algorithms (e.g., neural networks, logistic regression, etc.), which are described in Section II.

a) When the training dataset is ready, we choose a suitable ML algorithm to build the data model. It is possible to evaluate the complexity of the data model by conducting an in-depth analysis of the data characteristics [32]; for instance, what sort of ML model ought to be used for the training dataset? to what extent should the data information be described by the model structure in terms of complexity? is a single model structure adequate on its own? or it requires a multiple model structure? and so forth. In regression-based modeling, (*i*) a non-linear learning algorithm, such as a neural network or support vector machine model, is needed for strong non-linear correlations; (*ii*) for linear attribute relationships, a multivariate linear regression learning method can be selected; (*iii*) a multiple regression model

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structure must be used if numerous conditions have been found in the training dataset. Since there isn't a single, effective technique for selecting models, the ML community offers a number of helpful guidelines or criteria, such as the *Akaike Information Criterion*, *Bayesian Information Criterion*, *Hannan-Quinn Criterion*, etc. [32,37].

b) Using a ML algorithm on the training dataset, model parameters can be determined once the model structure has been selected. The model performance must be evaluated using a variety of techniques, including cross-validation, parameter sensitivity analysis, model stability analysis, and robust analysis before it is put to use online [32]. An additional separate testing or validation dataset is needed to verify the model and its performance evaluation. On the other hand, there are situations in which we might not be able to gather enough information for independent model validation and performance evaluation. The Bayesian learning algorithms and other re-sampling techniques like bagging and boosting can be applied in this situation.

As it can be noticed by the contributions of this Section, just like our ML structures realize multiple-step programs rather than single-step procedures (see Section I), with a similar analogy, our reference architecture moves the vision towards a *system-oriented point-of-view* rather than a *framework-oriented point-of-view*. Overall, the latter should be intended as a relevant innovation in big data systems research.

IV. A Reference Architecture for Big OLAP Data Cubes over Clouds

In this Section, we present the complementary reference architecture that serves to address the related issues of managing and mining big OLAP data cubes in Cloud environments. Particularly, this Section presents and describes the representation of big OLAP data cubes over Clouds via a multi-column-based representation and the paradigm of compressing the deriving multi-column representations.

When OLAP data cubes must be processed and managed on top of Cloud architectures, which are essentially *distributed* in nature, they convey in the so-called big OLAP data cubes. Since these data structures tend to easily become massive in size (e.g., [38]), they cannot be naturally processed and managed via centralized architectures, and even traditional experiences devoted to achieve effective and efficient distributed representations of such structures (e.g., [39]) fail. The reason for that is the need for different approaches in multidimensional data representation over Clouds, thus realizing a real research innovation. To be convinced of this, focus on traditional MOLAP-based representation of OLAP data cubes (e.g., [2]). According to this approach, OLAP data cubes are represented as *multidimensional arrays*, which, of course, are very hard to be implemented on top of distributed (data) repositories, especially for their *indexing data structure requirements* (e.g., [40]). On the other hand, as regards distributed aggregates computing, it should be noted that some aggregate operators are suitable to be computed according to this model (e.g., SUM, COUNT), whereas some others are not (e.g., AVG).

Following this main conceptual and methodological motivation, we propose an innovative mechanism for representing big OLAP data cubes over Clouds, which is inspired by previous research experiences in the context of *in-memory database management* (e.g., [41]). According to this

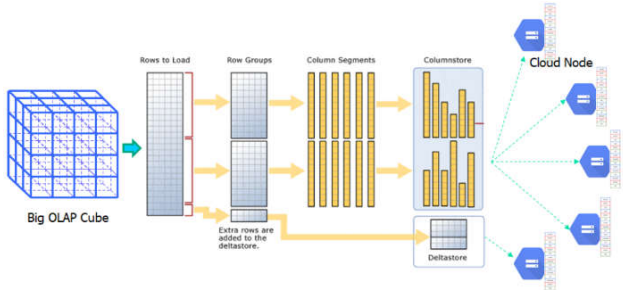

Fig. 2. A Reference Architecture for Big OLAP Data Cubes over Clouds

mechanism, big OLAP data cubes are first *partitioned into Columns*, and then these columns are *distributed across Cloud nodes* of the reference (Cloud) architecture. This Cloud-based, column-based representation model is fully compliant with modern big data processing platforms. For instance, consider the case of integration with Apache Spark: it is worthy to notice that, here, one whole (OLAP) column or partitions of it can be easily processed by ad-hoc Apache Spark libraries, even according to a full in-memory processing schema (e.g., for evaluating the COUNT aggregate of the elements of the column that satisfy a given filtering condition). This contributes to increase the performance of the entire big data framework. Fig. 2 shows our reference architecture for big OLAP data cubes over Clouds.

Despite the proposed reference architecture, big OLAP data cubes over Clouds can still pose severe limitations for what regards their accessing, querying, and mining phases (e.g., [42,43]). Therefore, we propose to further enrich our reference architecture via well-understood *data compression paradigms*. Data compression allows us to effectively and efficiently achieve a significant speed-up while ensuring the *accuracy* of the underlying analytical process (e.g., [44]). Data compression has been applied in the specific context of OLAP data cube compression (e.g., [45]), with alternate fortune. Here, we propose to still follow this general paradigm, and make use of data compression methods to further enhance the computational capabilities of our reference architecture for big OLAP data cubes over Clouds.

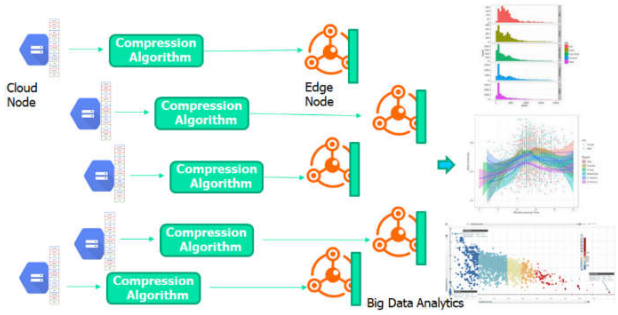

Fig. 3. Big OLAP Data Cube Compression over Clouds

As shown in Fig. 3, after the columns derived from the column-based representation of the input big OLAP data cube are distributed over the Cloud nodes of the target architecture. After that, they are compressed in order to gain into a more efficient data representation that is still compliant with the goals of the target multidimensional big data analytics process, where decimal precision is not mandatory (e.g., [46]). The compressed representations of columns are obtained via ad-hoc *compression algorithms* specifically tailored to multidimensional (i.e., OLAP) data. Then, the compressed columns are delivered on top of ad-hoc *edge nodes* of the

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overall Cloud architecture to achieve a proper and original *big data architecture*, particularly devised for big OLAP data cubes over Clouds. It should be noted that this approach follows some recent paradigms that appeared in literature recently (e.g., [47,48]). The final (multidimensional) big data analytics processing is, therefore, issued on the compressed columns, the latter scheme is perfectly suitable for a large number of real-life big data applications. To become convinced of this, it suffices to consider application scenarios such as social network analysis, massive graph data processing, scientific and statistical distributed data repositories, and so forth.

How to compress multidimensional data columns? This is a critical question in our investigated context. While in literature there exists a plethora of proposals in the data compression research area, in our specific application scenario, we seek for data compression techniques particularly suitable to multidimensional and OLAP data since the latter expose specific characteristics (e.g., [49]). This is an open question of our research, which deserves future investigations.

V. Conclusions and Future Work

This paper has presented an innovative approach for supporting machine-learning-based multidimensional big data analytics within Cloud environments through the utilization of multi-column OLAP data cube compression. Specifically, we initiated our discussion by articulating the importance and motivations behind the integration of ML into multidimensional big data analytics within Cloud systems. Additionally, we propose a reference architecture that details the working methodology of these innovative ML structures over big data, as well as the introduction of a complementary reference architecture for supporting the management of the proposed framework over Clouds via column-based big OLAP data cube compression paradigms. Beyond its immediate applications, this paper extends its impact by unveiling novel research opportunities and disruptive technologies, thus paving the way for the research community to achieve new milestones in the ever-evolving field of multidimensional big data analytics.

Future work is mainly oriented towards making our framework more compliant with emerging and innovative research challenges dictated by *modern big data trends* (e.g., [50-55]).

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